Exploration Slows Preference Capture Without Changing Which Preferences Are Reached

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Abstract

That recommenders move the preferences they estimate is established. How far a system moves someone has been measured as amplification, as drift, and as distance from an unmanipulated counterfactual. We propose a measure that asks instead where the user ends up, and compare the two readings. A user agent with a latent taste vector selects among K surfaced items, exposure moves the taste toward what was consumed, and a curator re-estimates from recent consumption. Across three conditions ($n = 40$ runs, 300 timesteps), the closed engagement loop more than doubles drift against an open-loop control (0.680 against 0.283, $p = 0.0001$). The endpoint reading separates the conditions differently. Under the closed loop the user's final taste sits at cosine 0.908 to the curator's estimate from timestep 20 and 0.320 to the user's own starting point, while under the open loop the two alignments are indistinguishable (0.718 and 0.717). Adding 20% forced random exploration reduces drift to 0.494 while alignment with the curator's early estimate remains at 0.924. A sweep over exposure strength and selection concentration finds the early estimate nearer than the origin in all 12 cells, with alignment confined to 0.846–0.935, while the drift gap ranges over $+0.647$ down to -0.388 and reverses sign under strong exposure. The two readings therefore disagree, and they disagree most where capture is most complete. Distance-based auditing returns its most reassuring values in exactly those conditions.

1 Introduction

A recommender system estimates what someone likes, and the only evidence available is what that person engaged with. Engagement was measured over a slate the system itself selected. The estimate is therefore fit to behaviour the estimator shaped, and nobody in the loop observes what the person would have preferred over a slate they never saw.

Preferences move under repeated exposure. Zajonc [1968] established that repeated exposure to a stimulus enhances attitude toward it with no reinforcement required. Place a preference that moves under exposure inside a system that selects the exposure, and the question of whose taste the system serves stops having an obvious answer.

Several lines of work establish that recommenders move preferences. Jiang et al. [2019] give a theoretical analysis of degenerate feedback loops in which system decisions influence user beliefs and preferences, which then shape the feedback the system receives, and they separate echo chamber effects from filter bubble effects. Kalimeris et al. [2021] model preference amplification in a matrix factorisation recommender, with users drifting toward recommended content while the system adapts to the updated preferences, and they evaluate mitigation strategies on a production video recommender. Coppolillo et al. [2025] build a simulation framework for the same setting and introduce metrics for quantifying drift in user preferences over time. Our model is a simplified instance of the dynamics all three describe, and we claim no novelty for the dynamics themselves.

Carroll et al. [2022] addresses the normative question directly. They argue that a system trained by long-horizon optimisation has an incentive to shift user preferences toward states that are easier to satisfy, and they propose estimating induced shifts before deployment and constraining policies to a trust region of shifts deemed safe. Their safety criterion is defined against how users would shift without system interference. Ashton and Franklin [2022] argue that any such criterion requires knowledge of meta-preferences if it is to respect user autonomy.

Chaney et al. [2018] supply the comparison our design uses. They show that training on algorithmically confounded data homogenises user behaviour relative to an open-loop randomised platform. Their users hold fixed underlying preferences, so harm is measured as behaviour diverging from what an unconfounded system would have delivered. In our model the preferences themselves move and no fixed true preference remains to diverge from.

Against this work our contribution is narrow. The preference-shift literature measures how far a system moves a user, whether as amplification [Kalimeris et al., 2021], as drift [Coppolillo et al., 2025], or as distance from an unmanipulated counterfactual [Carroll et al., 2022]. We measure something else: how close the endpoint is to the curator's own early estimate, which requires no counterfactual and no notion of an undisturbed baseline. We then show that the distance-based reading and the endpoint-based reading disagree, and that they disagree most where capture is most complete.

A second line of work concerns cultural markets. Salganik et al. [2006] show that social influence increases both inequality and unpredictability of outcomes, with quality only partly determining success. Their result is across worlds: run the market again and different songs win. Ours concerns a single run.

We contribute an endpoint-based measure of preference capture that requires no counterfactual, evidence that forced exploration changes the distance a user moves while leaving the endpoint measure unchanged, and a sweep showing that the two readings disagree in sign under strong exposure.

2 Method

2.1 Scope of the operationalization

This is a simulation. No user data was collected and no deployed system was measured. Users are single vectors rather than people with contexts and lives outside the feed; items are points rather than works; and one curator stands in for what is usually a stack of ranking and business constraints. The claim under test concerns the behaviour of a mechanism, which surfaces K items, observes a selection, re-estimates and repeats, under stated assumptions.

2.2 The loop

An item pool of 2,000 items is drawn uniformly on the unit sphere in $D = 8$ dimensions. A user holds taste p, initialised at p_0 uniformly at random. A curator holds an estimate $\hat{p}$, initialised at random and therefore carrying no information about the user at $t = 0$.

At each of $T = 300$ timesteps the curator surfaces $K = 10$ items. The user selects among the surfaced set, and only that set, by softmax on utility with concentration $\beta = 6$:

$$P(select x_j) \propto \exp(\beta\, x_j \cdot p)\,, \qquad x_j \in surfaced.$$

Consumption then moves taste, $p \leftarrow \mathrm{unit}(p + \alpha x)$ with $\alpha = 0.05$, and the curator re-estimates as the unit mean of the last 30 consumed items.

2.3 The exposure update (disclosed)

The exposure update produces every result below. We take it from the mere-exposure literature [Zajonc, 1968] rather than inventing it, and we implement it as a fixed linear step with no habituation, satiation or reactance. All three are documented and all three would reduce the effect. A reader who holds that adult aesthetic preferences are stable under exposure is describing a model in which none of this occurs, and should read the results as conditional on that assumption.

2.4 Conditions

The three conditions appear in Table 1.

Table 1: The three conditions. All share the item pool, selection rule, exposure update, estimation window and random seeds, and differ only in how items are surfaced.

Condition	Surfacing rule
`random`	K items drawn uniformly from the pool. Open loop: the curator's estimate never affects what the user sees.
`engagement`	Top K by $x \cdot \hat{p}$. The closed loop.
`engagement_explore`	Top $K - 2$ by estimate, plus 2 drawn uniformly from outside the top set.

The `random` condition is the control the argument depends on. Taste moves under any surfacing rule, because the exposure update applies to every consumed item regardless of how it was selected. Without an open-loop comparison, drift under engagement would establish only that exposure changes preferences, which is Zajonc's finding rather than ours. The contrast isolates what closing the loop adds.

2.5 Measures

Drift. $1 - \cos(p_t, p_0)$, the distance between the user's taste and its starting value. This is the quantity a platform can compute from its own logs.

Authorship. At the end of a run we compute $\cos(p_T, p_0)$, alignment with the user's own origin, against $\cos(p_T, \hat{p}_{20})$, alignment with the curator's estimate from timestep 20. Timestep 20 is chosen because it is early: the curator has observed 20 selections out of 300 and its estimate is still close to its random initialisation. If the user's final taste is nearer to that early estimate than to their own starting point, the estimate anticipated the destination.

Surfaced homogeneity. Mean pairwise cosine among the surfaced set, a descriptive check on whether the slate narrows.

Confidence intervals are bootstrap percentile intervals over runs, with the run as the independent unit. Comparisons use two-sided permutation tests on the difference in means [Efron and Tibshirani, 1993].

3 Results

3.1 Authorship (headline)

Table 2: Main results, $n = 40$ runs per condition, 300 timesteps each; brackets are bootstrap 95% CIs. Under both engagement conditions the user's final taste is far nearer the curator's timestep-20 estimate than the user's own starting point.

Condition	Final drift	$\cos(p_T, p_0)$	$\cos(p_T, \hat{p}_{20})$
`random`	0.283 [0.239, 0.333]	0.717 [0.667, 0.762]	0.718 [0.660, 0.771]
`engagement`	0.680 [0.604, 0.758]	0.320 [0.243, 0.397]	**0.908** [0.878, 0.933]
`engagement_explore`	0.494 [0.436, 0.558]	0.506 [0.445, 0.564]	**0.924** [0.900, 0.945]

The closed loop more than doubles drift against the open-loop control (Table 2, Figure 1a): 0.680 (95% CI 0.604–0.758) against 0.283 (0.239–0.333), difference $+0.397$, $p = 0.0001$, both $n = 40$.

The authorship columns carry the finding, and Figure 1c shows the same contrast. Under `random` the two alignments are indistinguishable, 0.717 and 0.718, which is what an uninformative comparison

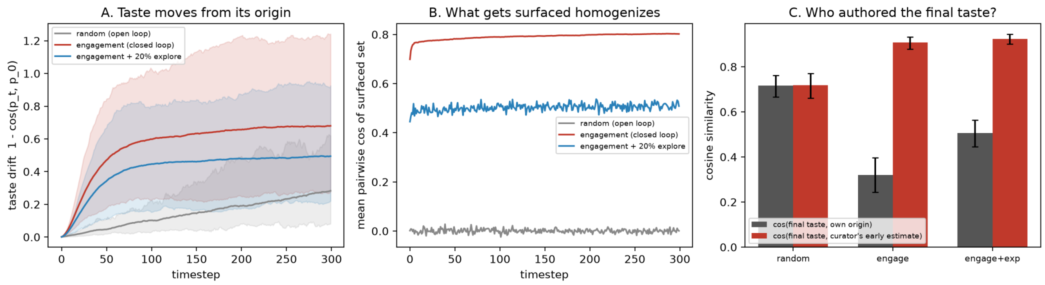

Figure 1: (a) taste drift $1 - \cos(p_t, p_0)$ over 300 timesteps, with shaded bands giving percentile intervals across runs. (b) mean pairwise cosine similarity among the surfaced set. (c) final alignment of the user's taste with its own origin (grey) against alignment with the curator's timestep-20 estimate (red), by condition, with bootstrap 95% CIs over $n = 40$ runs. The two bars are indistinguishable under open-loop surfacing and separate under both engagement conditions.

should produce: a curator whose estimate never reaches the slate describes the user about as well as the user's own history does. Under `engagement` they separate. The user's final taste sits at 0.908 to the curator's estimate from timestep 20 and 0.320 to the user's own starting point.

3.2 Exploration reduces drift without changing authorship

Adding 20% forced exploration reduces drift from 0.680 to 0.494, with non-overlapping intervals, and moves alignment with the user's origin from 0.320 to 0.506. On the quantity a platform can compute, the intervention works.

Alignment with the curator's early estimate is 0.924 (0.900–0.945) with exploration against 0.908 (0.878–0.933) without it. The intervals overlap almost entirely and the point estimate is nominally higher with exploration.

Forced exploration is the standard response to filter-bubble dynamics and it does change how far the user moves. The randomness is inserted into the slate, and the slate is still assembled around an estimate the slate produced.

3.3 Slates narrow under the closed loop

Mean pairwise cosine similarity among surfaced items rises over a run under both engagement conditions and remains flat under `random` (Figure 1b). We report this as a descriptive check rather than a finding. It confirms that a top-K rule does what a top-K rule mechanically must do, and it is not independent evidence for the authorship result.

3.4 The drift metric reverses sign where capture is strongest

We swept exposure strength $\alpha \in \{0.02, 0.05, 0.10, 0.20\}$ against selection concentration $\beta \in \{3, 6, 12\}$ at $n = 20$ runs per cell, running both `engagement` and `random` in each.

The early estimate is nearer the final taste than the origin is in all 12 cells. Alignment with the early estimate stays within 0.846 to 0.935 across the grid.

The drift gap does not behave this way. It ranges from $+0.647$ at $(\alpha = 0.02, \beta = 3)$ to -0.388 at $(\alpha = 0.20, \beta = 6)$, and it is negative in 4 of 12 cells, all at $\alpha \geq 0.10$. At $\alpha = 0.20$ the closed loop produces less movement than random surfacing. At $(\alpha = 0.20, \beta = 3)$ alignment with the user's origin reaches -0.084, so the final taste is weakly anti-correlated with where the user started.

The mechanism is worth stating directly. Under strong exposure the user is captured early, and a captured user stops moving: the items surfaced already align with the current taste, so the exposure update has little effect. An analyst measuring drift in those cells would conclude that the engagement loop preserves the user's taste, because the user is barely changing. The authorship measure gives the opposite reading, and gives it most strongly in the cells where drift gives the weakest.

3.5 A reproducibility null

The reported runs were produced on a different machine from the one used during development. The `random` condition agrees to four decimal places across the two. The engagement conditions differ by up to 0.003, roughly two percent of the width of the confidence intervals. The engagement path selects items with `argsort` over a 2,000-element matrix product, so small floating-point differences between numerical libraries occasionally reorder the top-K set, and those differences compound over 300 timesteps. The `random` path selects without reference to that product and is unaffected. Seeded runs are therefore bit-identical within a machine and approximately equal across machines, and no reported comparison changes under the discrepancy.

4 Discussion

Drift is the natural quantity to instrument. It is cheap to compute from logs and available to any platform. It also reverses sign in the cells where capture is most complete, so an audit built on how far user preferences moved will return its most reassuring numbers under the conditions that warrant the most concern.

Forced exploration is widely deployed and it addresses the quantity it was designed to address. It does not change authorship, because it operates inside a slate still assembled around a self-generated estimate. An intervention that addressed authorship would have to break the dependence of the estimate on the estimator's own past output, which is a larger change than adding randomness to a ranking.

Nothing in this model deceives the user. They see items, they select freely among them, and their selections are recorded accurately. What they never have is the counterfactual: the slate they were not shown, and the preference they would have reached under it. Agency here is exercised at every timestep inside a selection set someone else assembled, and the aggregate of freely made selections is a destination the curator's estimate anticipated at timestep 20.

5 Limitations

Simulation. No user data was collected and the connection to deployed systems is an argument. **The exposure update.** Taste moving toward consumed items is the assumption every result rests on. It is grounded in the mere-exposure literature but implemented as a fixed linear step with no habituation, satiation or reactance. **Positioning against prior work.** The dynamics we simulate are established in prior work [Jiang et al., 2019, Kalimeris et al., 2021, Coppolillo et al., 2025]. We claim novelty only for the endpoint measure and the disagreement between the two readings. Our characterisation of the prior measures rests on published descriptions rather than reimplementation. If any of that work already contains an endpoint-based measure of the kind used here, the contribution reduces to the sign reversal of §3.4. **Single curator, single objective.** Real systems balance engagement against diversity and commercial constraints, so the pure-engagement curator here is a worst case rather than a description. **Timestep 20 is a choice.** The authorship measure depends on when the early estimate is recorded. We selected 20 as early relative to 300 and did not test the sensitivity of the result to that selection. **Structural parameters unswept.** Item pool size, dimensionality, slate size K and the 30-item estimation window were held fixed; only α and β were varied. **No human validation.** Nothing here has been checked against how preferences move under deployed recommenders, which is the study this one motivates rather than replaces.

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NeurIPS Paper Checklist

1. **Claims**
   Answer: [Yes]
   Justification: The abstract and Section 1 state both headline numbers (final drift 0.680 against 0.283, $p = 0.0001$; alignment 0.908 with the curator's early estimate against 0.320 with the user's origin) and the exploration null. The sign reversal of the drift gap is stated in the abstract rather than reserved for the body.

2. **Limitations**
   Answer: [Yes]
   Justification: Section 5 covers the simulation-to-reality gap, the exposure update and the mechanisms omitted from it, the dependence of the related-work positioning on a reading of Chaney et al. [2018], the single-objective curator, the arbitrariness of the timestep-20 record, the structural parameters left unswept, and the absence of human validation.

3. **Theory assumptions and proofs**
   Answer: [N/A]
   Justification: The paper contains no theorems.

4. **Experimental result reproducibility**
   Answer: [Yes]
   Justification: Section 2 gives the item pool, dimensionality, slate size, selection rule and concentration, exposure rate, estimation window, timestep count and all three conditions. Section 3.5 reports the observed cross-machine agreement and its cause.

5. **Open access to data and code**
   Answer: [Yes]
   Justification: The simulation, the sweep, per-run result files and the figure-generating code are available at `https://github.com/AIscend-Research/whose-taste-is-it`. The code is two self-contained Python files depending only on NumPy and Matplotlib. Seeds are fixed; Section 3.5 reports the observed cross-machine agreement and its cause.

6. **Experimental setting/details**
   Answer: [Yes]
   Justification: Sections 2.2–2.5 specify every parameter and both swept ranges.

7. **Experiment statistical significance**
   Answer: [Yes]
   Justification: Every reported mean carries a bootstrap percentile 95% CI over runs ($n = 40$ per condition, the run as the independent unit), and comparisons use two-sided permutation tests with exact p-values. The exploration null in §3.2 is reported as a null with its overlapping intervals stated.

8. **Experiments compute resources**
   Answer: [Yes]
   Justification: CPU-only agent-based simulation with no model training. The full set of reported runs including the sweep completes in under five minutes on a laptop.

9. **Code of ethics**
   Answer: [Yes]
   Justification: No personal data, no human subjects, no scraped material and no dual-use risk. All agents are simulated.

10. **Broader impacts**
    Answer: [Yes]
    Justification: Section 4 argues that drift-based audits of recommender systems return their most reassuring values under the conditions of most complete capture, and that forced exploration does not address authorship. No artifact with foreseeable negative use is released.

11. **Safeguards**
    Answer: [N/A]
    Justification: No models, datasets or scraped material are released.

12. **Licenses for existing assets**
Answer: [N/A]
Justification: NumPy and Matplotlib only, under their respective open licenses.
13. **New assets**
Answer: [N/A]
Justification: See item 5.
14. **Crowdsourcing and research with human subjects**
Answer: [N/A]
Justification: The study involves no human subjects.
15. **Institutional review board (IRB) approvals**
Answer: [N/A]
Justification: No human subjects research was conducted.
16. **Declaration of LLM usage**
Answer: [Yes]
Justification: No LLM is a component of the method; the study is agent-based simulation over synthetic vectors. We declare usage for transparency. An LLM assistant was used substantially during implementation and drafting, including writing simulation code, constructing the sweep and drafting prose. The research questions, the choice of the endpoint measure, and all interpretation are the authors'. The reported runs were executed by an author, and every numeric claim in this paper was checked against the released result files.